Tectonic Stress and the Spectra of Seismic Shear Waves from Earthquakes

JAMES N. BRUNE

Institute of Geophysics and Planetary Physics
University of California, San Diego
Scripps Institution of Oceanography, La Jolla, California 92037

An earthquake model is derived by considering the effective stress available to accelerate the sides of the fault. The model describes near- and far-field displacement-time functions and spectra and includes the effect of fractional stress drop. It successfully explains the near- and far-field spectra observed for earthquakes and indicates that effective stresses are of the order of 100 bars. For this stress, the estimated upper limit of near-fault particle velocity is 100 cm/sec, and the estimated upper limit for accelerations is approximately $2g$ at 10 Hz and proportionally lower for lower frequencies. The near field displacement u is approximately given by $u(t) = (\sigma/\mu)\ \beta\tau(1 - e^{-t/\tau})$ where σ is the effective stress, μ is the rigidity, β is the shear wave velocity, and τ is of the order of the dimension of the fault divided by the shear-wave velocity. The corresponding spectrum is

$$\Omega(\omega) = \frac{\sigma\beta}{\mu}\frac{1}{\omega(\omega^2 + \tau^{-2})^{1/2}}$$

The rms average far-field spectrum is given by

$$\langle\Omega(\omega)\rangle = \langle\mathcal{R}_{\theta\phi}\rangle\frac{\sigma\beta}{\mu}\frac{r}{R}\ F(\epsilon)\frac{1}{\omega^2 + \alpha^2}$$

where $\langle\mathcal{R}_{\theta\phi}\rangle$ is the rms average of the radiation pattern; r is the radius of an equivalent circular dislocation surface; R is the distance; $F(\epsilon) = \{[2 - 2\epsilon][1 - \cos(1.21\ \epsilon\omega/\alpha)] + \epsilon^2\}^{1/2}$; ϵ is the fraction of stress drop; and $\alpha = 2.21\ \beta/r$. The rms spectrum falls off as $(\omega/\alpha)^{-2}$ at very high frequencies. For values of ω/α between 1 and 10 the rms spectrum falls off as $(\omega/\alpha)^{-1}$ for $\epsilon < \sim 0.1$. At low frequencies the spectrum reduces to the spectrum for a double-couple point source of appropriate moment. Effective stress, stress drop and source dimensions may be estimated by comparing observed seismic spectra with the theoretical spectra.

Dislocation models have been successfully used in several studies of long-period waves and static deformations caused by earthquakes [*Knopoff*, 1958; *Maruyama*, 1963; *Burridge and Knopoff*, 1964; *Haskell*, 1964; *Press*, 1965; *Savage and Hastie*, 1966; *Aki*, 1966; *Brune and Allen*, 1967; *Wyss and Brune*, 1968; *Berckhemer and Jacob*, 1968]. Recent theoretical studies on source mechanism have been given by *Archambeau* [1968] and *Burridge* [1969].[1] Two recent studies [*Aki*, 1968; *Haskell*, 1969], have successfully applied dislocation theory to the study of near-source displacements. Unfortunately the dislocation models used have been somewhat arbitrary in the specification of the time function of the dislocation motion. Aki arbitrarily assumed the dislocation occurred as a step function in time, whereas Haskell assumed a ramp function with slope estimated from the expected duration of fault slippage. In this study, the time function is related directly to the effective stress available to accelerate the two sides of the fault. The results provide a physical basis for integration methods such as those of Aki and Haskell and in addition provide a basis for understanding the time function and spectrum at high frequencies, which are of particular interest in engineering seismology and in the study of

[1] I wish to thank one of the reviewers of my paper for pointing out the paper by Burridge. Burridge has used a model similar to that assumed here and has presented numerical and analytical solutions for velocity and displacement (in the plane of the fault) for a two dimensional model. His results support the approximate near source solutions given in this paper.

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earthquake spectra. Obvious modifications of the idealized model considered here may be made in the study of particular earthquakes.

Near-Source Model of an Earthquake Dislocation

We model an earthquake dislocation as a tangential stress pulse applied to the interior of a dislocation surface; in particular, as a stress pulse in one direction applied to the fault block on one side of the fault, and in the opposite direction to the other fault block (see Figure 1). We assume that the pulse is applied instantaneously over the fault surface, i.e., we neglect fault propagation effects. This assumption is discussed at the end of this section. During rupture, the fault surface is equivalent to a surface unable to transmit shear waves from one block to the other, i.e., the fault surface during rupture is totally reflecting to shear waves. Initially, the motion at a point near the center of the fault occurs as if the fault plane were infinite (before the end effects can propagate to the center of the fault). The stress pulse sends a pure shear stress wave propagating perpendicular to the dislocation surface. The initial time function for this pulse follows directly from the boundary conditions

$$\sigma(x, t) = \sigma H(t - x/\beta) \tag{1}$$

where $H(t)$ is the Heaviside unit step function

$$H(t) = 0 \qquad t < 0$$

$$H(t) = 1 \qquad t > 0$$

σ is the effective shear stress and β is the shear wave velocity.

The tangential displacement u corresponding to (1) may be obtained by integration since $\sigma = \mu \partial u/\partial x$, giving at $x = 0$

$$\begin{aligned} u &= 0 & t &< 0 \\ u &= (\sigma/\mu)\beta t & t &> 0 \end{aligned} \tag{2}$$

The stress pulse propagates with shear velocity β perpendicular to the dislocation surface. For a point near the fault, trace displacement increases linearly with time until the effects of the boundaries of the dislocation at distance r reach the observation point and stop the linear increase in displacement. Thus, for an observation point near the center of the dislocation

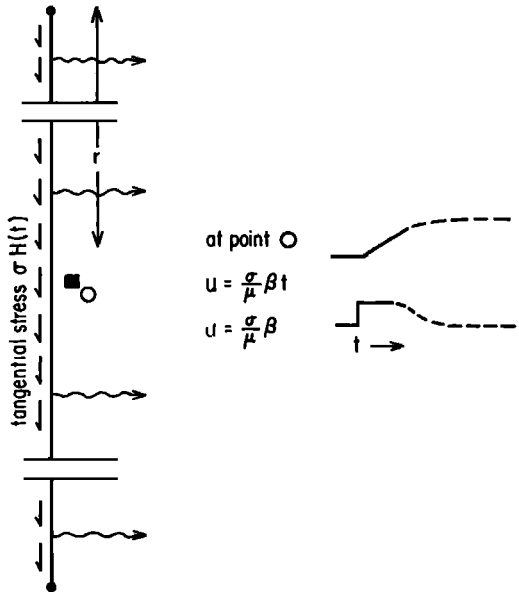

Fig. 1. Illustration of the stress-pulse model for an observation point, O, near the dislocation surface (fault plane). In this schematic diagram the displacement u is tangential to the dislocation surface. A shear pulse on the inner right surface of the dislocation sends a shear wave propagating to the right (plus x direction). A wave of opposite sense propagates to the left. r is the approximate radius of an equivalent circular dislocation.

surface, the particle displacement is initially given by (2). Near the edges of the fault the boundary conditions require compression and rarefaction of the medium, with consequent motions normal to the fault plane. These normal motions are comparable in magnitude to the tangential motion near the center of the fault but produce angular momentum of opposite sign, so that the net angular momentum is zero. These normal motions correspond to the other couple of the equivalent double-couple source representation. The initial stress has no angular moment since the stress is applied in one plane (i.e., there is no moment arm).

The spectrum of (2) is

$$\Omega(\omega) = \int_0^\infty \frac{\sigma}{\mu} \beta t \, e^{-i\omega t} \, dt = -\frac{1}{\omega^2} \frac{\sigma\beta}{\mu} \tag{3}$$

The initial particle velocity is:

$$\dot{u} = \frac{\sigma}{\mu} \beta \tag{4}$$

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For $\sigma = 100$ bars $= 10^8$ dynes/cm^2, $\mu = 3 \times 10^{11}$ dynes/cm^2, and $\beta = 3$ km/sec $= 3 \times 10^5$ cm/sec, (4) gives $\dot{u} = 100$ cm/sec. A stress of 1000 bars gives $\dot{u} = 1000$ cm/sec. 100 cm/sec appears to be a good value for the upper limit of initial velocities observed in earthquakes (Table 1).

The highest value of $\dot{u}$ observed to date is 76 cm/sec for the Parkfield, California, earthquake of June 28, 1966. [*Housner and Trifunac*, 1967]. In this case, the strong-motion seismograph was located *nearly* on the fault trace. The El Centro, California, earthquake of May, 1940, generated velocities of 43 cm/sec on the E–W component and 41 cm/sec on the N–S component. The vectorially resolved velocities were probably somewhat higher. These observations indicate that the effective stresses operating during large earthquakes are of the order of 100 bars, in good agreement with estimates from static dislocation studies [*Chinnery*, 1964; *Brune and Allen*, 1967], with an estimate based on the lack of an observable heat-flow anomaly along the San Andreas fault in California [*Brune et al.*, 1969], and with the apparent stress for large earthquakes obtained by dividing the energy by the seismic moment [*Aki*, 1966; *Brune*, 1968].

In reality the stress pulse will be applied as a propagating source, and the spectrum will be modified by a propagation directivity function defined later. The particle velocities, $\dot{u}$, are always much smaller than the rupture propagation velocity v (since the stresses are much less than the shear modulus μ), and thus the interaction of particle motion velocity with the rupture velocity will always be small, i.e., the average energy spectrum for instantaneous application of stress will be approximately valid; however, the finite velocity of rupture will focus the energy in the direction of propagation and thus introduce a strong azimuthal dependence upon the radiation.

Maximum Near-Source Acceleration

Although the accelerations given by (2) are infinite at the arrival time of the pulse, the forces remain finite because the accelerating masses are zero in the limit. We may predict the maximum accelerations expected at any finite frequency or for any frequency band from (3). This will correspond to the maximum accelerations expected for masses with volume of the order of the cube of the wavelength. During cracking of rocks, rockbursts, etc., extremely high accelerations are obtained, but only at very high frequencies. For engineering seismology, these high accelerations at high frequencies are not important since the building resistance to damage at high frequencies is very great. The frequencies of greatest interest in engineering seismology are less than 10 Hz,

TABLE 1. Maximum Velocities Observed Near Earthquakes

Earthquake	x comp., cm/sec	y comp., cm/sec	Date	M	Δ (Epicenter w.r.t. Strong-Motion Station)
	E–W	N–S			
Port Hueneme	9.1	12.7	March 18, 1957	4.7	~4 miles southeast
	N 65°E				
Parkfield Sta. 2	76.2	···	June 27, 1966	5.6	~20 miles northwest
	E–W	N–S		[(6.7), 7.0],	
El Centro	43	41	May 18, 1940	6.4	~7 miles southeast
	E–W	N–S			
El Centro	19.1	43.2	Dec. 30, 1934	6.5	~35 miles south
	N 80°E	N 10°W			
Olympia	22.9	20.3	April 13, 1949	7.1	~45 miles north-northwest
	N 69°W	S 21°W			
Taft	22.9	12.7	July 21, 1952	7.7	~40 miles east-southeast

Note.
Data from D. Hudson and M. O. Trifunac (personal communication).

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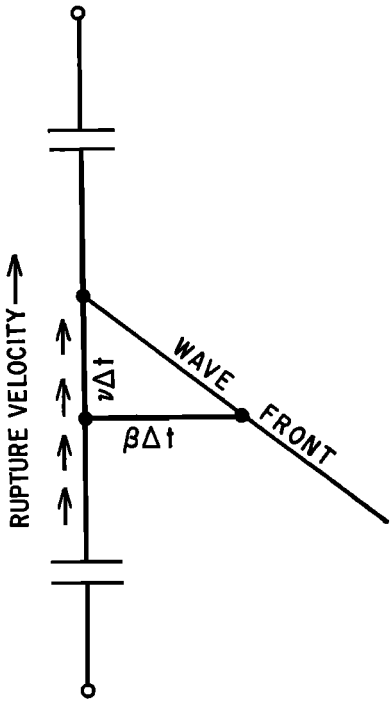

Fig. 2. Illustration of the finite rate of stress application for a traveling rupture.

and, in fact, higher frequencies are generally not strongly recorded on strong-motion seismographs (D. Hudson, personal communication).

Consider a small time interval Δt as the rupture propagates along the fault plane with velocity v (Figure 2). A shear pulse travels away from the fault with velocity β. A small mass of dimensions one unit of length deep, $v\Delta t$ along the fault and approximately[2] $\beta\Delta\tau$ perpendicular to the fault, is accelerated by a force $\sigma\cdot 1\cdot v\Delta t$ applied at the fault surface. The acceleration is then given by

$$\ddot{u} = \text{force/mass} = 2\sigma/(\rho\beta\Delta t) \qquad (5)$$

In the limit of small Δt, $\ddot{u}$ approaches ∞; however, for any finite mass and finite Δt (finite frequency), $\ddot{u}$ is finite. For example, if we consider a volume 0.3 km in dimension ($\Delta t = 0.1$, frequency $\sim$ 10 Hz) and $\sigma = 100$ bars, we have

$$\ddot{u}(10\text{ Hz}) \cong \sigma 2\cdot 10^{-5} = 2g \qquad (6)$$

[2] For $v < \beta$ the wave front actually proceeds ahead of the rupture, and the accelerated mass is slightly larger.

Alternatively, we may estimate the maximum accelerations by considering the contribution of finite band of frequencies from 0 to some cutoff angular frequency, ω_s.

$$u(t) = \frac{1}{2\pi}\frac{\sigma}{\mu}\beta\int_{-\omega_s}^{\omega_s} -\frac{1}{\omega^2}e^{i\omega t}\,d\omega \qquad (7)$$

$$\begin{aligned}\ddot{u}(t) &= \frac{1}{2\pi}\frac{\sigma}{\mu}\beta\int_{-\omega_s}^{\omega_s} e^{i\omega t}\,d\omega \\ &= \frac{1}{\pi}\frac{\sigma}{\mu}\beta\omega_s\left(\frac{\sin\omega_s t}{\omega_s t}\right)\end{aligned} \qquad (8)$$

For a cutoff frequency of 10 Hz and $\sigma = 100$ angular frequency, ω_s.

$$\ddot{u}(10\text{ Hz}) \simeq 2g \qquad (9)$$

in agreement with (6).

At any finite distance from the fault, the effect of a finite rupture velocity will limit the high-frequency spectrum. For example, as a rough approximation, at an observation point a small distance x from the dislocation surface the maximum contribution to the displacement comes from the nearest segment of the fault, of length about x, over which a stress pulse propagating with velocity v is applied. The spectrum $\Omega(\omega)$ is modified by the propagation directivity factor $(\sin\chi)/\chi$ [*Ben-Menahem*, 1961], where $\chi = (\omega x/2\beta)(\beta/v - \cos\theta_0)$. Since θ_0 is about $\pi/2$, $\chi \cong \omega x/2v$. For values of χ greater than about $\pi/2$, $(\sin\chi)/\chi$ oscillates with mean amplitude, decreasing as $1/\chi$. Thus the high-frequency acceleration will be reduced for frequencies above approximately $(\omega x)/(2v) \cong \pi/2$ or $\omega \cong \pi v/x$. If we consider an observation point at a distance $x = 0.2$ km from the dislocation surface, with $v = 3$ km/sec, this gives $\omega \cong 45$ or $f \cong 8$ Hz. In this case, assuming a cutoff frequency of 10 Hz as in (9) will be a good approximation. Very near the dislocation surface we may expect higher accelerations at high frequencies.

In applying the calculations in this section to estimate maximum likely accelerations, two additional factors must be taken into account: focusing and amplification of energy, and the effect of the depth to the region of maximum stress release. Focusing and amplification (by transition to a medium with lower rigidity) may locally increase the velocities and accelerations;

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on the other hand, the fact that the depth to the zone of high stress release will usually be greater than one kilometer will tend to reduce the observed accelerations.

Richter [1958] reports a few cases of effects in the epicentral regions of large earthquakes that indicate accelerations have locally exceeded the force of gravity. The largest acceleration observed to date on a strong-motion seismograph is 0.6 g for the Koyna, India, earthquake of 1967. The Parkfield, California, earthquake of June 28, 1966, produced an acceleration of about 0.5 g and the El Centro, California, earthquake of May, 1940, an acceleration of 0.3 g. All these high accelerations occurred in the frequency range 5 to 10 Hz (D. Hudson, personal communication).

We conclude that accelerations produced by large earthquakes may be expected to locally exceed the acceleration of gravity for frequencies near 10 Hz. On the other hand, the observation that accelerations exceeding gravity have seldom occurred suggests that the effective stresses operating along earthquake faults must be of the order of 100 bars. An effective stress of 1000 bars would be expected to occasionally produce local accelerations of over 10 g at 10 Hz.

Maximum Local Magnitude M_L for Earthquakes

The high-frequency near-source spectrum is independent of the source size and directly related to the effective stress. An instrument recording directly on the fault surface will record approximately the same high-frequency spectrum regardless of the dimensions of the rupture surface. Thus the local earthquake magnitude scale will have an upper limit corresponding to the case where a standard seismometer happens to be located directly on the fault trace.

The local earthquake magnitude M_L defined in *Richter* [1958] is

$$M_L = \mathrm{Log}_{10} A - \mathrm{Log}_{10} A_0 \qquad (10)$$

where A is the amplitude, in millimeters, recorded on the standard Wood-Anderson torsion seismometer (free period, 0.8 sec; high frequency magnification, 2800), and A_0 is the empirically determined amplitude corresponding to a magnitude zero earthquake. A_0 is a function of distance and is tabulated in *Richter*, [1958]. The high-frequency response of the torsion seismometer extends to about 40 Hz, but a variety of factors usually limit the observed frequencies to about 10 Hz. The combined effect of the ω^{-2} source effect and the ω^2 long-period response of the instrument give, on the seismogram, approximately a constant spectral density versus frequency up to about the free period of the instrument, 0.8 sec. The record displacement in the near field is thus given approximately by

$$\begin{aligned} u &\cong \frac{2800}{2}\frac{\sigma}{\mu}\beta\int_{-\omega_0}^{\omega_0} e^{i\omega_0 t}\,d\omega \\ &\cong \frac{2800}{\pi}\frac{\sigma}{\mu}\beta\omega_0\left(\frac{\sin\omega_0 t}{\omega_0 t}\right) \end{aligned} \qquad (11)$$

At $t = 0$ this gives, for $\sigma = 100$ bars and $\omega_0 = 0.8$ sec, $u_{max} \cong 6 \times 10^5$ mm $-\mathrm{Log}\, A_0 = 1.4$ at very near distances. The maximum magnitude is thus

$$M_L(\max) = 5.8 + 1.4 = 7.2 \qquad (12)$$

It is difficult to measure local magnitudes for large earthquakes because the ordinary Wood-Anderson seismographs are off scale for deflections greater than about 300 mm. After the 1952 Tehachapi earthquake drove most of the Wood-Anderson torsion seismometers in Southern California off scale, Richter put into operation several torsion seismometers with gains of 100 and 4 (shape of response same as for the Wood-Anderson), but unfortunately no very large earthquakes ($M > 7$) have occurred in California since then. To the author's knowledge, the greatest local earthquake magnitude determinations in California, based solely on records of Wood-Anderson torsion seismometers, at near distances, are about 6.5 (El Centro, 1940 [*Trifunac and Brune*, 1970]; Borrego Mountain, 1968 as reported by Richter, personal communication). A reduced gain Wood-Anderson torsion seismometer recording at the site of the accelerometer, which gave the 0.5 g acceleration near Parkfield, would have given an equivalent trace deflection of about 3×10^5 mm (1000 is the approximate gain of the Wood-Anderson at 1-sec period, the approximate period of the rise of the 30-cm displacement pulse). This corresponds to a local magnitude of about

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$$M_L = 5.4 + 1.4 = 6.8$$

which is 0.4 less than the estimated upper limit for M_L given above. This calculation also suggests that the effective stresses operating during earthquakes in Southern California are of the order of 100 bars and not as high as 1000 bars.

Fault Friction and Effective Stress

Following *Orowan* [1960], if we assume that during the fault rupture a frictional stress σ_f acts to resist the fault slippage, then it follows that when, as a result of slippage, the available stress across the fault has decreased to σ_f, the fault will lock itself and slippage will cease. At the initiation of slippage the effective stress will be $\sigma_0 - \sigma_f$; an amount of energy $\sigma_f \cdot u$ per unit area will be lost as frictional heating, and an amount of energy $\frac{1}{2}(\sigma_0 - \sigma_f)u$ per unit area will go into the generation of seismic waves. Thus the dynamics of the rupture behave as if there were no friction and the available stress were only $(\sigma_0 - \sigma_f)$. For this reason the stresses deduced from seismic waves refer only to the effective stress $\sigma = (\sigma_0 - \sigma_f)$. Although there is no way to determine the stress σ_f (and consequently σ_0) from seismic waves, in the case of the San Andreas fault we may estimate an upper limit for σ_f from the lack of an observable heat-flow anomaly centered over the fault. *Brune et al.* [1969] concluded that available heat-flow data indicate an upper limit for σ_f of the order of a hundred bars. σ_f may actually be very small for large earthquakes because the energy available is sufficient to melt the rock along the fault zone, thus reducing the friction to nearly zero. We assume that creep occurring after the rapid fault slippage has stopped has a time constant too great to generate observable seismic waves.

Near-Field Effect of the Finite Dislocation Size

As the effects of the edges of the dislocation surface become felt at the observation point, the particle velocity will be decreased and approach zero for times large compared to the distance to the edge divided by the shear velocity β. This effect may be approximated by replacing (2) by

$$u(x = 0, t) = (\sigma/\mu)\,\beta\tau(1 - e^{-t/\tau}) \qquad (13)$$

$$\dot{u}(x = 0, t) = (\sigma/\mu)\,\beta\, e^{-t/\tau} \qquad (14)$$

At $t = 0$, (14) gives $\dot{u} = (\sigma/\mu)\beta$ as in (4).

The Fourier transform of (13) gives

$$\Omega(\omega) = \frac{\sigma}{\mu}\beta\omega^{-1}(\omega^2 + (\tau)^{-2})^{-1/2} \qquad (15)$$

In this approximation

$$\tau = 0(r/\beta) \qquad (16)$$

This case is diagrammatically illustrated in Figure 3. The high-frequency spectrum and the initial rise velocity are not altered, but the velocity decays to zero. As slippage proceeds, perpendicular displacements develop with approximately the same magnitude as the tangential displacements [*Aki*, 1968; *Haskell*, 1969].

Near-Field Effect of Fractional Stress Drop

If the effective stress does not drop to zero but is stopped at some fraction ϵ of complete stress drop, the rise-time and high-frequency spectra are not drastically changed, but the long-period behavior and spectra are reduced by ϵ. This effect is considered in a later section for far-field radiation. The spectrum is multiplied by a function $F(\epsilon) = [(2 - 2\epsilon)(1 - \cos 1.21\epsilon) + \epsilon^2]^{1/2}$. The effect in the near field is to produce about $1.6/\epsilon$ higher velocities and accelerations at high frequencies than would be expected for 100% fractional stress drop corresponding to the same dislocation.

Dislocation During the Parkfield Earthquake

Aki [1968] *and Haskell* [1969] concluded that the dislocation at the time of the Parkfield earthquake must have been about 60 cm and 90 cm, respectively, more than twice the value of the displacement pulse observed (30 cm) and much larger than the several-cm offset observed at the surface after the earthquake [*Smith and Wyss*, 1968]. The model given here supports that conclusion. Unless some unusual amplification occurs, the displacement pulse at any distance from the displacement surface never exceeds one-half the dislocation value. We may estimate the minimum source size for the Parkfield displacement pulse by using our model. If the velocity is about 60 cm/sec, as observed for the Parkfield earthquake, then the time duration necessary to produce a dis-

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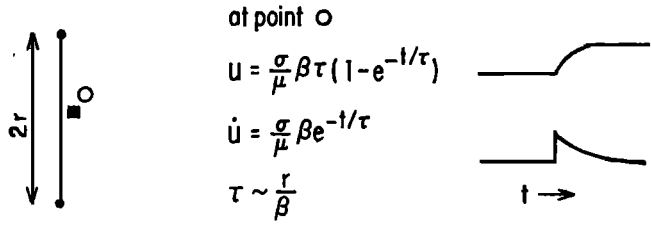

Fig. 3. Illustration of the effect of finite source dimensions on the near-field displacement.

placement of 30 cm is ½ sec. From equation 16, this corresponds to a value of τ of about ½ sec, giving a minimum dimension of r of approximately 1 km. Thus a large dislocation over a relatively small dislocation surface nearby can explain the large displacement pulse observed at Parkfield. In view of the often complicated pattern of energy release observed in earthquakes, it is not a safe assumption that this large value of dislocation was consistent over the complete fault surface. It is possible, of course, that anomalous amplification and focusing of energy by a nonlinear effect in the sediments near the strong-motion seismograph may have contributed to the large displacement observed.

Far-Field Radiation Spectrum

As the distance to the observation point increases, diffraction reduces the long-period or static spectrum, i.e., the static field decays at a higher order of distance R than the dynamic field. At large distances and large wavelengths (compared to the source dimensions), the effect of the opposing side of the fault diffracts around the dislocatiion surface and differentiates the far-field spectrum [*Keilis-Borok*, 1960]. This gives the long-period, large-distance equivalent source as a double couple [*Maruyama*, 1963; *Burridge and Knopoff*, 1964].

To study the average far-field spectrum, we approximate the effect of diffraction by multiplying the displacement function by an exponential with decay time of the order of r/β and multiply by a factor $f \cdot r/R$ to take into account spherical spreading:

$$u = f \cdot (r/R)(\sigma/\mu)\beta t'' e^{-\alpha t''} \tag{17}$$

where

$$t'' = t - R/\beta \tag{18}$$

this gives, at $t'' = 0$

$$\dot{u} = f(r/R)(\sigma/\mu)\beta \tag{19}$$

The Fourier transform of (17) is

$$\Omega(\omega) = f(r/R)(\sigma\beta/\mu)\frac{1}{\omega^2 + \alpha^2} \tag{20}$$

The function (17) is similar in form to the approximate far-field function found by *Jeffreys* [1931*a*, *b*] for the case of a stress pulse applied to the interior of a spherical surface, as given by *Bullen* [1963, p. 76]:

$$u\phi = \frac{2\sigma r^2}{(3)^{1/2}\mu R}\sin\frac{(3)^{1/2}\beta t''}{2r}e^{-3\beta t''/2r} \tag{21}$$

The initial rise of (21) can be obtained by expanding the sine term for small values of the argument:

$$u_\phi \sim \frac{\sigma}{\mu}\beta\frac{r}{R}t''e^{-3\beta t''/2r} \tag{22}$$

in agreement with the form of (17).

As an approximation for the average pulse from a circular dislocation, we shall choose f and α such that the spectrum in the long-period limit agrees with the dislocation source moment, and in the high-frequency limit, conserves the energy-density flux at large distances.

Immediately after the application of the stress pulse, the energy is flowing away from the dislocation surface, and its high-frequency spectrum is given by (3). The total area over which the energy flux occurs is 2A since an equal amount of energy flows from the opposing surface. This energy passes through a spherical surface at large distance R in the form of both P waves and S waves. The area of the surface is $4\pi R^2$. Thus the conservation of energy density at high frequencies gives

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$$S \cdot 2A \cdot \left(\frac{\sigma\beta}{\mu\omega^2}\right)^2 = 4\pi R^2 \langle \Omega_s^2 \rangle \tag{23}$$

where S is a factor correcting for the conversion of S waves to P waves and $\langle \Omega_s^2 \rangle$ is the average squared spectral density of S waves averaged over the spherical surface. This gives, for a circular dislocation with $A = \pi r^2$,

$$\begin{aligned} \langle \Omega_s^2 \rangle^{1/2} &= \left(\frac{S}{2}\right)^{1/2} \frac{r}{R} \frac{\sigma}{\mu} \frac{\beta}{\omega^2} \\ &= (0.4)^{1/2} \Omega_s^{(\max)} \end{aligned} \tag{24}$$

since for a double couple with radiation pattern $R_{\theta\phi}$:

$$\frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \, R_{\theta\phi}^2 = 0.4 \tag{25}$$

[Wu, 1966]. Thus for f in (17) we find

$$f = (S/0.8)^{1/2} \tag{26}$$

We now determine a condition for f and α from the requirement that in the long-period limit the spectral density agree with that for a double couple of the same moment as that from the dislocation. Following *Keilis-Borok* [1960] the far-field S-wave radiation for a double couple is given by:

$$\Omega_S(\omega) = \mathcal{R}_{\theta\phi} M_0 (4\pi\rho R\beta^3)^{-1} \tag{27}$$

where $\mathcal{R}_{\theta\phi}$ is the radiation pattern, and M_0 is the moment of one couple of a double couple source.

From *Aki* [1966] and *Keilis-Borok* [1959] we also have, for $\lambda = \mu$,

$$M_0 = \mu A \bar{u}_d \tag{28}$$

$$A = \pi r^2 \tag{29}$$

$$\bar{u}_d = \frac{3}{4} u_d^{(\max)} = \frac{\sigma}{\mu} r \frac{18}{7\pi} \tag{30}$$

where $u_d^{(\max)}$ is the maximum final dislocation for 100% stress drop. Hence

$$M_0^{100} = \sigma r^3 (18/7) \tag{31}$$

In the long-period limit, the spectrum of (20) becomes

$$\Omega_S(\omega) = \mathcal{R}_{\theta,\phi} \frac{\sigma}{\mu} \beta f(r/R)(1/\alpha^2) \tag{32}$$

Equations 27–32 give

$$f = \frac{9}{14\pi} \left(\frac{\alpha r}{\beta}\right)^2 \tag{33}$$

From (33) and (26) we obtain the following condition for S and α:

$$\alpha r/\beta = (S/0.8)^{1/4} (14\pi/9)^{1/2} \tag{34}$$

For $S = 0.9$, this gives $\alpha = 2.28\beta/r$ and $f = 1.06$; for $S = 0.8$, $\alpha = 2.21\beta/r$ and $f = 1.00$. As a sufficient approximation we assume

$$f = 1 \tag{35}$$

$$\alpha = [(14\pi)^{1/2}/3]\beta/r = 2.21\,\beta/r \tag{36}$$

The displacement and corresponding spectrum are:

$$u(\mathcal{R}, t, \theta, \phi) = \mathcal{R}_{\theta\phi} \frac{\sigma}{\mu} \beta (r/R) t'' \, e^{-2.21(\beta/r)} t'' \tag{37}$$

$$\Omega_S(\mathcal{R}, \omega, \theta, \phi) = \mathcal{R}_{\theta\phi} \frac{\sigma}{\mu} \beta \frac{r}{R} \frac{1}{\omega^2 + (2.21\,\beta/r)^2} \tag{38}$$

Figure 4 gives a plot comparing the function (37) with Jeffreys's solution for a stress pulse on the interior of a sphere. As expected, the decay time is a little shorter for a circular dislocation and the total area under the curve is less, corresponding to the lower moment of the source. The total radiated S wave energy in this model represented by equation 32 is:

$$E_S = \frac{\sigma^2 \beta \pi r^2 f^2 (0.4)}{\mu\alpha} \tag{39}$$

This is 44% of the total energy available from the dislocation, ($\frac{1}{2}\sigma\langle \mathrm{Au}_d \rangle$). The model does not specify the total P-wave energy, but it should probably be assumed to be less than that for S waves. Hence somewhat less than 88% of the total available energy is accounted for, and the average spectral densities are probably about 10% too low, in the neighborhood of $\omega/\alpha = 1$. This discrepancy arises from the assumed simple form of the far-field displacement (equation 17 and Figure 4). It is small compared with present uncertainties in the measurement of spectra.

Stick Slip and Fractional Stress Drop

In many studies of earthquakes the stress drop appears to be only a small fraction of the

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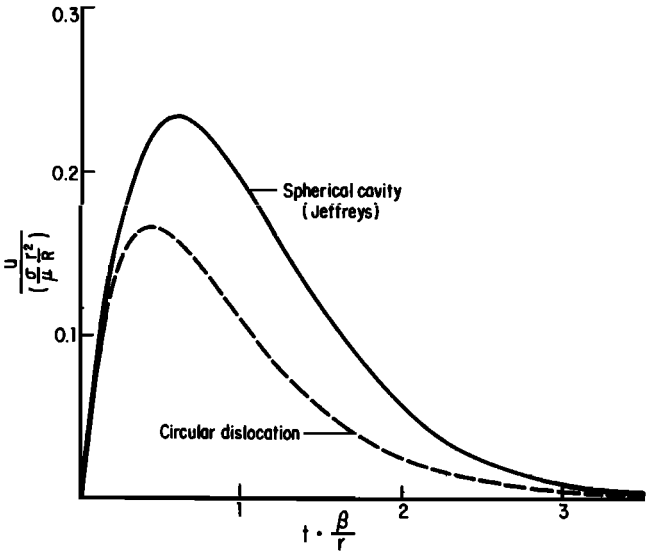

Fig. 4. Comparison of theoretical far-field pulse shapes. Curves for Jeffreys's model of a stress pulse on the inside of a sphere and the circular dislocation model developed here are shown.

effective stress [*Aki*, 1967; *Brune and Allen*, 1967; *Wyss and Brune*, 1968; *King and Knopoff*, 1968]. A similar effect is observed in laboratory experiments and is called 'stick slip' [*Brace and Byerlee*, 1966]. We can model this effect by supposing that at a short time t_d after the initial shear stress is applied a reverse stress of $1 - \epsilon$ is applied. The long-period spectra and seismic moment will be reduced to ϵ times the value for 100% stress drop; however, the very high frequency spectra will be much less affected. Several mechanisms can lead to average slip (or average stress drop) over the fault plane being less than that corresponding to 100% effective stress drop. For example, if the rupture propagates along the fault plane and the fault plane locks itself after the rupture has passed, the rupture may travel large distances (not determined by the effective stress) with the total effective stress operating only for a short time as the rupture passes a given point. A similar situation occurs if the stress release is not uniform and coherent over the whole fault plane, or if the rupture proceeds as a series of multiple events. The dimensions of the zone of energy release will not be determined by the effective stress, and even though the total effective stress will be acting over certain portions of the fault plane for short periods of time, the average slip (stress drop) over the fault plane is much less than that corresponding to 100% effective stress drop.

The displacement (37) is modified as follows; neglecting the factors r/R and $\mathcal{R}_{\theta\phi}$,

$$\mu_\epsilon(t) = \frac{\sigma}{\mu}\beta t'' e^{-\alpha t''} - (1 - \epsilon)$$

$$\frac{\sigma}{\mu}\beta(t'' - t_d)e^{-\alpha(t''-t_d)} \qquad t > t_d \quad (40)$$

t_d is approximately given by ϵ times one-half the 100% stress-drop dislocation divided by the initial particle velocity, i.e.,

$$t_d = \frac{1}{2}\frac{\epsilon\, u_d^{(\max)}}{\dot{u}} = \frac{2.21\epsilon}{\alpha}\frac{12}{7\pi} \simeq 1.21\,\frac{\epsilon}{\alpha} \quad (41)$$

Changing the exact value of t_d will change the position of interference nodes at high frequencies but will not critically change the average values of amplitude. Taking the Fourier transform of (40) we find that the spectrum is multiplied by a function $F(\epsilon) = |1 - (1 - \epsilon)e^{-i\omega t_d}|$ or

$$F(\epsilon) = \{[2 - 2\epsilon]$$

$$[1 - \cos(1.21\,\epsilon\omega/\alpha)] + \epsilon^2\}^{1/2} \quad (42)$$

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For large values of ω, $F(\epsilon)$ oscillates between ϵ and $2 - \epsilon$, with a mean value of

$$\epsilon + (2/\pi)^{1/2}(2 - 2\epsilon) \cong 1.6 - 0.6\epsilon \qquad (43)$$

In reality the dislocation may not stop abruptly, as assumed here, but may decelerate more gradually. This will reduce the high frequency spectrum somewhat over that given by (43). However, the dislocation must stop relatively abruptly if ϵ is small. For small values of ω, $F(\epsilon)$ approaches ϵ.

The Effect of Complex Rupture Propagation

In actual earthquake dislocations the effective stress will not be applied instantaneously over the dislocation but will be applied in a generally complex manner. The effect is to create a complex interference pattern in the far field. In the case of a smoothly propagating rupture, the interference pattern can be easily calculated [*Ben-Menahem*, 1961]. For the case of a moving point source, the interference pattern is given by the sin χ/χ function, where $\chi = (\pi b/cT)(c/v - \cos\theta_0)$, b is the length of rupture propagation, c is the phase velocity, T is the wave period, v is the velocity of rupture propagation, and θ_0 is the angle between the direction of propagation and the direction of the observation. The effect of a smoothly propagating source is to strongly focus the high-frequency energy in the direction of the rupture propagation, but since the same amount of energy is radiated as in the case of instantaneous application of stress, the rms spectral density taken over the radiation pattern will be the same. In a particular direction the spectrum may be considerably distorted from the case of instantaneous stress application. In this study we assume that the appropriate averaging over the radiation pattern has been performed or that corrections have been made for the effect of rupture propagation, so that the spectra may be interpreted in terms of the instantaneous stress application model.

Average Far-Field Spectrum

Combining all the effects considered above, we have the following relation for the rms far-field spectrum:

$$\langle\Omega(\omega)\rangle = \langle\mathcal{R}_{\theta\phi}\rangle \frac{\sigma\beta}{\mu}\frac{r}{R}F(\epsilon)\frac{1}{\omega^2 + \alpha^2} \qquad (44)$$

where $\langle R_{\theta\phi}\rangle$ is the rms average of the radiation pattern, 0.4. Figure 5 illustrates the properties of this function for various values of the fraction of stress drop ϵ. For small values of ω, the effect of rupture propagation becomes negligible; $F(\epsilon) \to \epsilon$, and $(\omega^2 + \alpha^2)^{-1} \to \alpha^{-2}$; hence, as $\omega \to 0$

$$\Omega(\omega) \to \mathcal{R}_{\theta\phi}\frac{\sigma\beta}{\mu}\frac{r}{R}\frac{\epsilon}{\alpha^2} \qquad (45)$$

or, since $M_0^{100} = (18/7)\sigma r^3$ and $\sigma^2 = 14\pi/9)(\beta/r)^2$:

$$\Omega(\omega) \to \mathcal{R}_{\theta\phi}(\epsilon\, M_0^{100}/4\pi\rho R\beta^3) \qquad (46)$$

in agreement with (27).

Thus the long-period spectrum is reduced to ϵ times its value for 100% fractional stress drop because the effective moment is ϵM_0^{100}.

For large values of ω, $F(\epsilon)$ oscillates with a mean value of $1.6 - 0.6\epsilon$, and $(\omega^2 + \alpha^2)^{-1} \to \omega^{-2}$. Hence as $\omega \to \infty$, the rms average spectrum becomes:

$$[\Omega(\omega)] \to \langle\mathcal{R}_{\theta\phi}\rangle \frac{\sigma\beta}{\mu}\frac{r}{R}(1.6 - 0.6\epsilon)\alpha^{-2}(\omega/\alpha)^{-2} \qquad (47)$$

where the square brackets indicate a mean value of amplitude through oscillations and nodes of amplitude of $F(\epsilon)$. Thus at high values of ω/α the rms spectrum falls off as $(\omega/\alpha)^{-2}$.

Summarizing the far field spectra, for low values of ω/α the rms spectrum approaches a constant of ϵ times the spectrum for 100% fractional stress drop. For values of ω/α near 1 the rms spectrum begins to fall off as $(\omega/\alpha)^{-1}$. For high values of ω/α the rms spectrum decays as $(\omega/\alpha)^{-2}$. For high values of ω/α the spectra have the slope of the ω^{-2} statistical model proposed by *Aki* [1968] rather than the ω^{-2} model proposed by *Haskell* [1964]. However, the function $F(\epsilon)$ causes the spectra to fall off only as $(\omega/\alpha)^{-1}$ until $\epsilon\omega/\alpha$ is about 1. Thus for $\epsilon = 0.1$ the $(\omega/\alpha)^{-2}$ falloff does not begin until ω/α is about 10, and for $\epsilon = 0.01$ it does not begin until ω/α is about 100. Thus, roughly speaking, the ω^{-2} model corresponds to large values of ϵ and the ω^{-1} model to small values of ϵ. It should be emphasized that for any particular direction of observation the spectrum will not have the form of the rms spectrum of equation 44 and Figure 5 because of the focusing effects

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of rupture propagation. One must take into account these effects either by averaging or by knowing the direction of rupture propagation. Both the corner frequency and the slope of the spectrum versus frequency will depend on azimuth.

Determination of Tectonic Stress from Far-Field Observations

By fitting the observed spectra (corrected for scattering and attenuation) to the spectra predicted by equation 44, we may solve for the effective stress σ, the source dimension r, and the fractional stress drop ϵ. For a given source dimension r, the spectrum at low frequencies (which may easily be determined from seismograms) is controlled by the effective seismic moment ϵM_0^{100}, whereas at high frequencies the spectrum is controlled by the effective stress σ. The factor $1.6 - 0.6\epsilon$ does not critically control the average level of the spectrum at high frequencies, the average level varying only from 1.6 to 1 as ϵ varies from 0 to 1. If the slippage does not stop abruptly as assumed here, the high-frequency spectra may be reduced somewhat. Physically, the spectrum at high frequencies is controlled by the energy density conservation equation 23 and is not critically dependent on the long-period behavior of the dislocation, i.e., on whether the stress finally drops 10% or 100% of σ.

Spectrum fitting to determine the absolute stress is equivalent to the method of using the ratio of seismic energy to seismic moment as suggested by *Aki* [1966] *and Brune* [1968]. The energy/moment ratio method of estimating stress applies the equation

$$\sigma = \mu E/\eta \, \epsilon \, M_0^{100} \tag{48}$$

where E is the radiated seismic energy and η is the efficiency of conversion of strain energy to seismic energy. This method corresponds to integrating the spectra given in this study to determine the total energy and then correcting for the seismic efficiency η.

The energy of earthquakes is very poorly known and is usually calculated by assuming an empirical energy-versus-magnitude relation such as that of *Gutenberg and Richter* [1956]. In practice the magnitude is usually estimated from a relatively narrow frequency band and does not accurately describe the energy radiated. Studies such as that of *Wyss and Brune* [1968] used the local earthquake magnitude, which probably gives a fair estimate of energy since it is defined by amplitudes recorded on the broad-band Wood-Anderson torsion seismograph. *Wyss* [1970] integrated body waves to estimate the energy and found fairly good agreement with values predicted by the Gutenberg-Richter energy relation. Spectrum fitting similar to that suggested here was also employed by *Berckhemer and Jacob* [1968] but for different source parameters. The curves in Figure 5 may be thought of as defining a 'scaling law' of seismic spectra based on the parameters stress, source dimensions and fractional stress drop. The scaling law of seismic

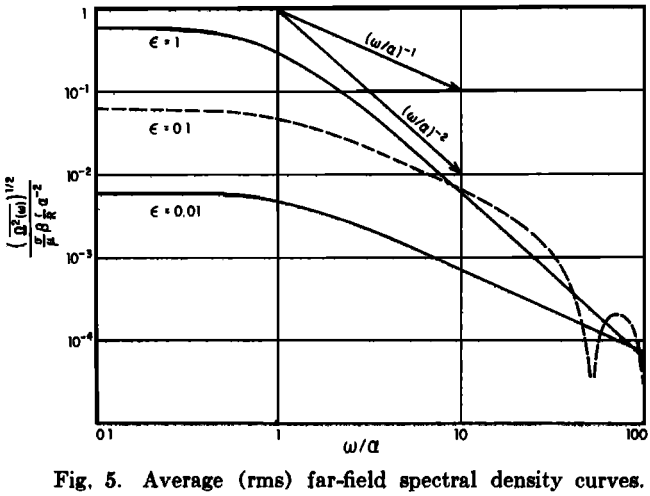

Fig. 5. Average (rms) far-field spectral density curves.

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spectra proposed by *Aki* [1968] was based on the magnitude and on a similarity assumption that does not take into account possible variations in fractional stress drop or regional variations in effective stress.

Conclusions

An earthquake model has been derived by considering the effective stress available to accelerate the sides of the fault. The model describes the near- and far-field displacement time function and spectrum and includes the effects of fractional stress drop. The model successfully explains near- and far-field spectra observed for earthquakes and indicates that the effective stresses operating during earthquakes are of the order of 100 bars. The results obtained in this study may be used to estimate effective stress, stress drop, and source dimensions by comparing observed seismic spectra with the theoretical spectra.

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